

COAL MINE SCREENING REPORT · King County, Washington

15500 Six Penny Ln

Newcastle, WA · Parcel 2724059013

Mapped workings under this parcel, at shallow depth

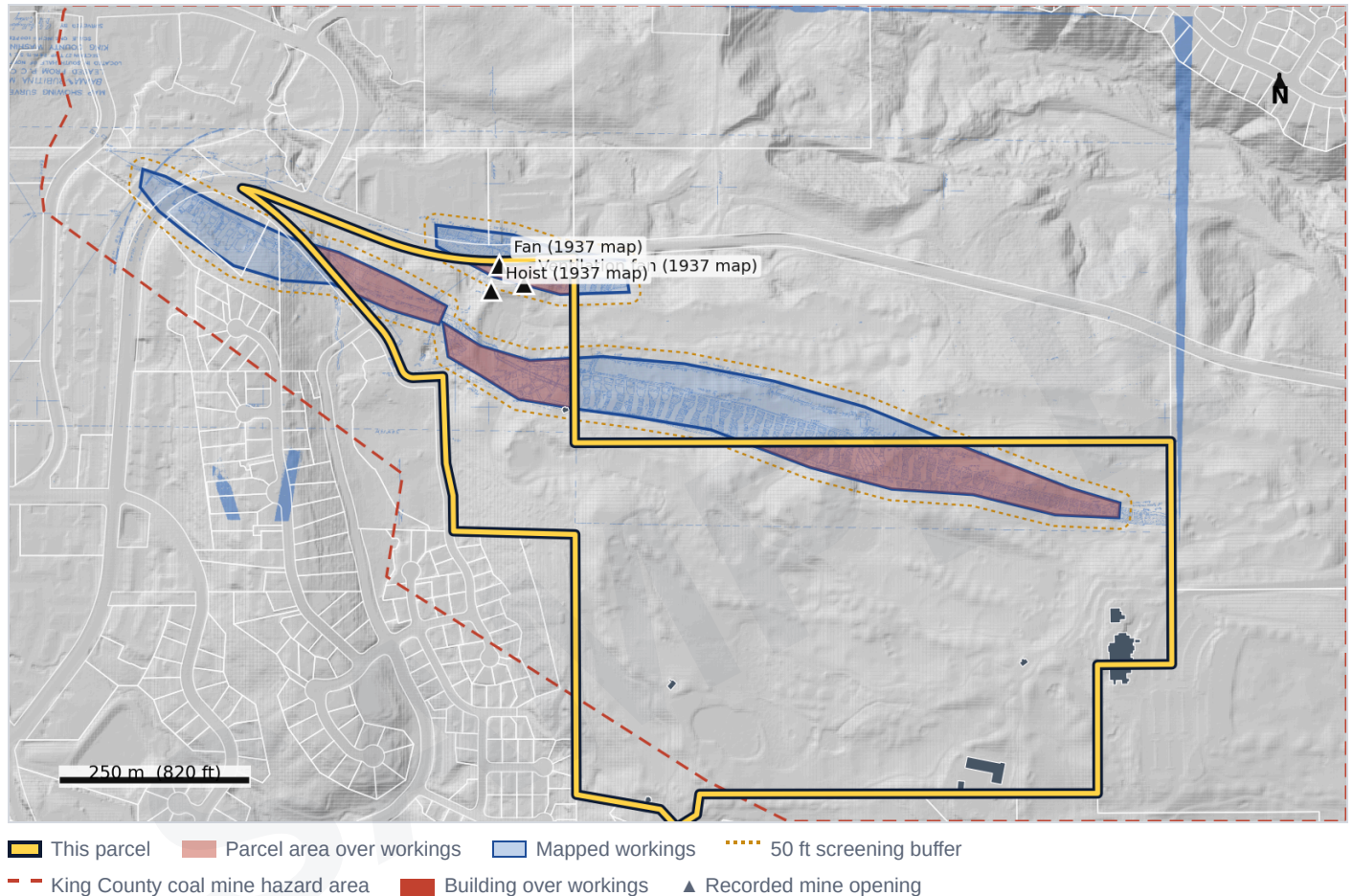
Source map: WA DNR K27_G, B.&R. (Baima & Rubitina) mine, 1937 survey

Parcel	2724059013 · 114.57 acres · Newcastle Golf Course
Under mapped workings	9% of the parcel (10.74 ac)
Within 50 ft of mapped workings	Yes
Depth to workings, undermined part	at or near the surface to 327 ft (median 107 ft)
Indicated hazard band (typical depths)	Severe 56% · Moderate 40% · Declassified 4%
King County coal mine hazard map (1990)	Parcel is inside the mapped area (96% of it)
Buildings on the parcel over workings	0 of 9 mapped buildings
Nearest recorded mine opening	Ventilation fan (1937 map), on the parcel

Under the depth bands the King County model and the City of Newcastle code use as *typical* guides (NMC 18.24.200: severe typically 0–150 ft, moderate 0–300 ft or cover-to-seam ratio under 10:1, declassified typically deeper than 300 ft), the undermined part of this parcel reads as **severe** at its shallowest. That is a screening indication from the map, not a classification: the class is set by a coal mine hazard assessment report prepared by a Washington-licensed engineering geologist or geotechnical engineer.

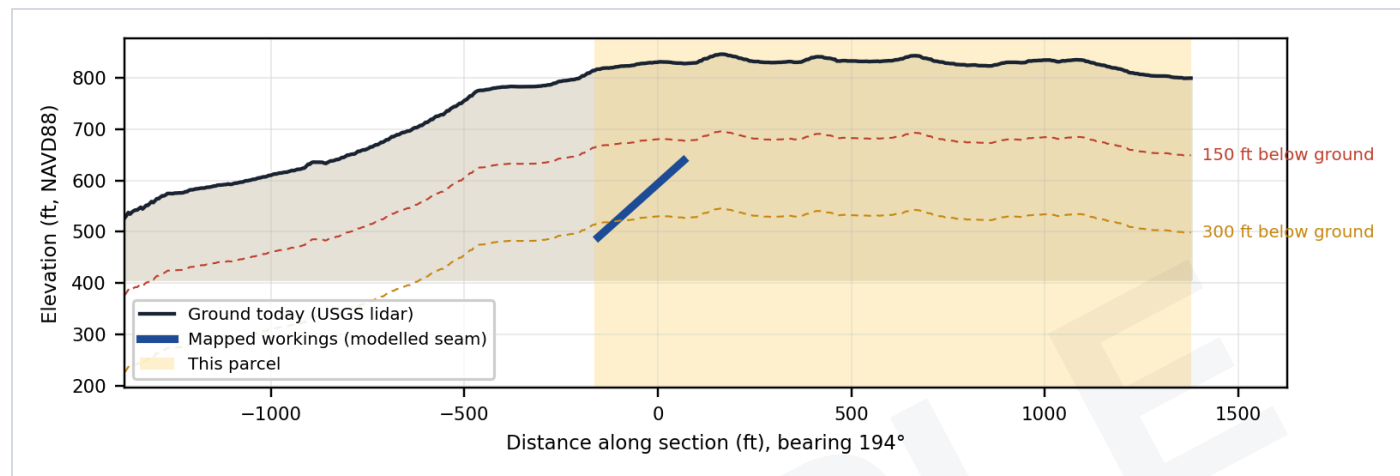
Public sample. Prepared by BIM Ops Studio from public records to show what a screening report contains. It was not commissioned by, and makes no statement about, the property owner.

Plan: the parcel over the mine



Background: USGS lidar hillshade with the 1937 mine sheet's linework (blue) laid on it. Parcel lines from King County.

Section through the parcel



Cut across the seam where it is steepest, through the undermined part of the parcel. Dashed lines show 150 ft and 300 ft below today's ground, the typical severe and moderate depths.

How deep the workings are

Depth was sampled at 1595 points over the undermined part of the parcel. In 18% of them the seam, extended at its recorded dip, would reach today's ground, so the workings there are treated as **at or near the surface** (the original map does not record their exact depth near the outcrop). Shallowest at or near the surface, median 107 ft, deepest 327 ft.

Where the map came from and how far to trust its position

Source map	WA DNR coal mine map K27_G: B.&R. (Baima & Rubitina) mine, Newcastle
Survey date	1937 (pillars noted as pulled 1937 on the No. 4 seam panel)
How it is placed on today's ground	WA DNR 2016 georeference of the sheet: 5 control points, their fit residual 0.56 ft. Residuals on so few points understate true error.
Our checks	Placement compared with today's streets and USGS lidar before the model was built (October 2026): consistent; no shift applied.
Depths come from	Gangway elevations written on the sheet (about 431-436 ft) and the 33-35 degree dips written in the chutes; the coal rises to the south.
Known gaps on this sheet	The west panel's gangway level is assumed equal to Lower Split #3 (not written). East of station 49 the sheet's elevations jump to 468-520 ft beside a note '#53 chute on Upper Seam': a possible second, higher seam that this model does not include. Near the outcrop the modelled seam would rise above today's ground; there it is treated as at or near the surface.

Historic mine maps placed from a few control points commonly carry position errors of tens of feet, so this report also flags anything within 50 ft of mapped workings. Mines are often larger than any single surviving map shows: unmapped or later workings may exist.

What this means for a project here

- Building on land underlain or affected by abandoned coal workings in King County and its cities requires a **coal mine hazard assessment report** by a Washington-licensed engineering geologist or geotechnical engineer before permits are issued (for example, NMC 18.24.120(B) and 18.24.208; KCC 21A.24).
- That report must delineate the severe, moderate and declassified areas on a survey map at not less than 1 inch = 200 feet. This screening report gives the engineer the georeferenced workings, depths and sources to start from.
- Because part of the parcel reads as **severe** at its shallowest, expect the assessment to include site investigation (for example borings or geophysics) in the undermined area.

What BIM Ops Studio can add

- **Interactive 3D** of this parcel over the mine, the route from the mine mouth and every building above the workings (live sample: bimopsstudio.com/mine-maps/3d/newcastle.html).
- Survey-scale workings drawings (DWG / PDF at 1" = 200' or larger) for the engineer of record.
- The same check across a portfolio, a corridor or every parcel in a development.

Method

The historic mine map is placed on today's coordinates, the mined-out outline is traced from it, and the seam is modelled at the elevations written on the map at the dips it records. Ground is the USGS 3DEP lidar surface. Depth to workings is ground elevation minus modelled seam elevation, sampled across the parcel. The parcel boundary and the 1990 coal mine hazard area come from King County GIS. The same model drives the interactive 3D view.

Sources

- WA DNR Coal Mine Map Inventory, sheet K27_G, and DNR 2016 georeferenced GeoTIFF
- USGS 3D Elevation Program (3DEP) lidar, 2 m
- King County GIS: parcels (PARCEL_ADDRESS_PUB_AREA), Sensitive Areas coal mine hazard (1990 SAO)
- OpenStreetMap buildings and streets
- City of Newcastle Municipal Code 18.24 (critical areas); King County Code 21A.24

Limitations

This is a desktop screening of public records. It is not a coal mine hazard assessment, a geotechnical report, an engineering or geological opinion, a survey, or a statement about the safety or value of the property. Positions, outlines and depths come from historic maps that may be incomplete or inaccurate, and from models built on them. Only a physical investigation by a licensed professional engineer or geologist can establish ground conditions. BIM Ops Studio provides mapping as a subconsultant and does not stamp hazard classifications.

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